



Full length article

Assessing the impacts of climate change on photovoltaic energy generation in Colombia using 3-hourly CMIP6 projections

Carlos G. Ramírez-Ruiz^{ID}*, Sergio Cabrales, Carlos F. Valencia

Industrial Engineering Department, University of los Andes, Cra 1 Este No 19 A-40 Bogotá, 111711, Colombia

ARTICLE INFO

Keywords:

Photovoltaic power potential
Climate change
Shared socioeconomic pathways
Statistical downscaling
CMIP6 models

ABSTRACT

Understanding how climate change will affect renewable energy systems is becoming increasingly important as countries accelerate their transition away from fossil fuels. Although photovoltaic (PV) energy is expected to be a key part of this transition, its long-term efficiency may be affected by changes in weather factors. In this study, we assess the impacts of climate change on PV energy generation in Colombia using statistically downscaled climate projections and a physically based PV panel model. Projections of weather-related variables (solar radiation and air temperature) are generated at 3-hourly resolution, providing a novel framework for understanding intra-day photovoltaic performance under future climate conditions. Future scenarios are based on three CMIP6 global climate models (GFDL-ESM4, MRI-ESM2-0, and BCC-CSM2-MR) and two emission pathways (SSP1-2.6 and SSP5-8.5). PV generation is estimated using a historical baseline (1994–2014) and compared with a mid-century period (2051–2060) to evaluate annual, monthly, and intra-day changes in PV potential. Results show spatially heterogeneous but overall modest changes in PV energy generation across Colombia, mainly driven by projected variations in solar radiation, while temperature-related efficiency losses have a smaller impact. Monthly analyses reveal substantial seasonal variations that are not apparent when considering annual averages alone. An assessment of temperature-only effects indicates that warming alone leads to small but consistent reductions in PV output under the SSP5-8.5 scenario.

1. Introduction

The rapid acceleration of global decarbonization has made renewable energy a key component of climate mitigation strategies [1]. Transforming the energy sector, which accounts for about one-third of human-made emissions, has become an international priority as greenhouse gas (GHG) emissions continue to drive global warming [2]. Solar photovoltaic (PV) systems have expanded rapidly due to efficiency improvements and significant cost reductions. Between 2010 and 2023, the prices of PV modules have decreased by 86%, making solar energy more competitive [3]. With nearly USD \$300 billion in global investment, solar PV has become the top recipient of renewable energy funding, establishing itself as a crucial part of worldwide decarbonization efforts [4].

Solar energy production is inherently sensitive to meteorological factors, especially surface solar radiation and air temperature [5]. Long-term changes in solar radiation have been observed worldwide, showing significant regional differences linked to aerosol loading, cloud cover, and shifts in atmospheric circulation [6]. For example, several regions have experienced “brightening” in recent decades—such as the United States, Europe, and China—linked to decreases in aerosols or

cloud cover [7,8], while others, including parts of South Asia and the Middle East, continue to show “dimming” trends because of ongoing aerosol emissions [9] despite advances in understanding these mechanisms, uncertainties persist regarding the roles of cloud–aerosol interactions and regional atmospheric processes, particularly in areas with limited in-situ observations are limited [10].

Earth System Models (ESMs) are vital tools for evaluating interactions among solar radiation, clouds, and aerosols, and for predicting future changes in solar resources [11]. Since the 1990s, the Coupled Model Intercomparison Project (CMIP) has coordinated global modeling efforts, producing standardized simulations that support most climate impact assessments. Earlier studies based on CMIP5 models have explored the potential impacts of climate change on PV generation across different regions, generally finding modest future decreases in PV potential, with regional exceptions linked to reductions in cloud cover [12,13]. More recent analyses using CMIP6 models show improved representations of cloud and aerosol processes and greater climate sensitivity in several models, leading to more detailed projections of solar radiation and air temperature [14,15]. Recent CMIP6-based studies suggest that future PV power potential will display significant

* Corresponding author.

E-mail address: cg.ramirez2513@uniandes.edu.co (C.G. Ramírez-Ruiz).<https://doi.org/10.1016/j.egycc.2026.100266>

Received 12 March 2026; Received in revised form 22 June 2026; Accepted 5 August 2026

Available online 17 August 2026

2666-2787/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

regional and seasonal variability, with projected increases in parts of Europe, South America, Australia, and eastern North America, and decreases in various regions of Asia and Africa [16]. Under low-emission scenarios, PV generation variability is expected to decrease in areas such as China [17], while air temperature and humidity are identified as primary drivers of a worldwide reduction in solar energy potential, with cloud and aerosol changes shaping the regional patterns of surface solar irradiance and PV output [18,19].

In South America, CMIP-based studies reveal substantial spatial heterogeneity in future PV potential. Zuluaga et al. [20] found that climate change may increase PV potential in northern Brazil by about 3% by mid-century due to reduced cloud cover and increased solar radiation, while temperature-induced efficiency decreases might cause an overall drop of around 5% across most of the country by the late 21st century. In a broader context, other research indicates that PV potential in the South American monsoon region could increase by roughly 1.5%–2.1% under different climate scenarios compared to historical conditions [21]. Narváez et al. estimated future changes in photovoltaic potential, showing regionally varied impacts by the end of the century, with decreases of around 15% and increases up to 7%, mainly due to changes in solar irradiance [22].

The proximity of Colombia to the equator results in high solar insolation, with an estimated photovoltaic solar energy generation potential of 32 GW [23]. Colombia has strong natural conditions for solar energy deployment, particularly in its northern coastal regions, with 4 to 12 h of sunshine per day and above-average solar radiation compared to global benchmarks [24]. This potential has driven rapid solar capacity expansion in recent years, reaching 1660 MW in 2025, representing 7.9% of the country's total installed capacity, with expansion plans targeting up to 7.92 GW [25]. This growth has also been fueled by Colombia's heavy reliance on hydropower and the need to reduce dependence on a single energy source that is vulnerable to climate variability. Historical and projected variations in temperature and precipitation patterns have increasingly threatened energy security, often requiring greater reliance on thermal generation sources such as natural gas and coal (Fig. A.13). In this context, the expansion of renewable energy sources (particularly photovoltaic) is essential to support a sustainable energy transition and enhance the resilience and reliability of the national electricity supply [26].

Colombia's pronounced geographic diversity and complex Andean topography result in strong spatial heterogeneity in PV potential, which national averages cannot adequately capture, highlighting the need for regionally detailed assessments. Recent studies indicate contrasting regional impacts: Narváez et al. reported relatively modest end-of-century changes in PV potential in southwestern Colombia (Nariño), with variations between –2.5% and +2.5% [27], while a CMIP6 multi-model ensemble analysis by Arregocés et al. projected much larger and spatially heterogeneous reductions at the national scale, with decreases of 10%–23% in the near future and up to 27%–35% by mid-to-late century in the Pacific, Andean, and Orinoquía regions, together with limited seasonal increases of 1%–4% in the Amazon region [28].

A key challenge in assessing the impacts of climate change on photovoltaic generation is the coarse spatial resolution of global climate models, which limits their capacity to represent local atmospheric processes, particularly in regions with complex topography such as the Colombian Andean mountains [29]. To address this limitation, downscaling techniques are commonly applied to translate large-scale climate signals into high-resolution, locally relevant projections. Statistical downscaling plays a central role in this process by enabling bias correction and by reconstructing the relationships between large-scale atmospheric predictors and local meteorological variables, which is essential for reliable regional PV assessments [30]. Although recent international studies have increasingly examined PV variability, stability, and low-generation events under future climate scenarios, analyses at sub-daily temporal scales remain relatively scarce. This gap is especially evident in Colombia, where detailed, high-resolution assessments of

climate change impacts on photovoltaic generation remain limited despite the rapid expansion of solar energy deployment.

This study examines how climate change affects PV energy production in Colombia using downscaled projections from CMIP6 global climate models. We consider two emission scenarios, SSP1-2.6 and SSP5-8.5, and employ a 3-hour resolution to better capture intra-day variations that are often overlooked in coarser analyses. Three global climate models (GCMs) are chosen based on their resolution and data availability: GFDL-ESM4, MRI-ESM2-0, and BCC-CSM2-MR. PV output is simulated through a physical panel model using projected solar radiation and air temperature, with results compared for a historical baseline (2005–2014) and a mid-century period (2051–2060). By combining high-resolution downscaled climate projections with PV modeling, this research offers the first comprehensive assessment of mid-century climate change impacts on photovoltaic generation in Colombia at a sub-daily resolution.

2. Materials and methods

Fig. 1 presents the methodological framework employed in this study. The approach consists of a sequential procedure designed to quantify the impacts of climate change on PV energy generation in Colombia. The analysis integrates multiple data sources, including technical parameters of PV panels, effective installed capacity of existing solar farms in Colombia, historical air temperature and solar radiation from the ERA5 reanalysis, and outputs from global climate models (GCMs) for both historical conditions and future climate Shared Socioeconomic Pathways (SSPs), particularly SSP1-2.6 and SSP5-8.5. Using these datasets, a statistical downscaling (SD) approach is first implemented to translate large-scale GCM information into locally relevant meteorological projections. An initial calibration and evaluation stage is carried out by comparing downscaled GCM outputs with ERA5 data over the historical period to identify the SD model that best reproduces observed temperature and radiation patterns. The selected SD model is then applied to GCM projections under the two climate change scenarios to generate future high-resolution meteorological time series. A multi-model ensemble is constructed using the GCMs, applying a genealogy-aware weighting scheme to account for both model performance and structural interdependence among CMIP6 models.

Finally, these downscaled projections are combined with PV panel characteristics and the effective capacity of Colombian solar farms within a PV performance model to estimate future PV energy generation and assess climate change impacts under low- and high-emission pathways. Projected changes were quantified as the difference between the future and historical periods, expressed in absolute terms for the meteorological variables (solar radiation and air temperature) and as percentage changes relative to historical PV power generation. The statistical significance of these changes was evaluated at the 95% confidence level using a Student's *t*-tests.

2.1. Study area

Colombia is located in the northwestern region of South America and lies along the equator. The country is administratively divided into 32 departments and has a population of approximately 53 million inhabitants [31]. Its territory is commonly grouped into five major natural regions with distinct geographic and climatic characteristics: the Caribbean region in the north along the Caribbean Sea; the Pacific region to the west, bordering the Pacific Ocean; the Andean region in the central part of the country, dominated by the three branches of the Andes mountain range; the Orinoquía region in the east, characterized by the Llanos plains within the Orinoco River basin near the Venezuelan border; and the Amazon region in the southeast, which forms part of the Amazon rainforest. This spatial aggregation is based on the official climatic regionalization defined by the Instituto Geográfico

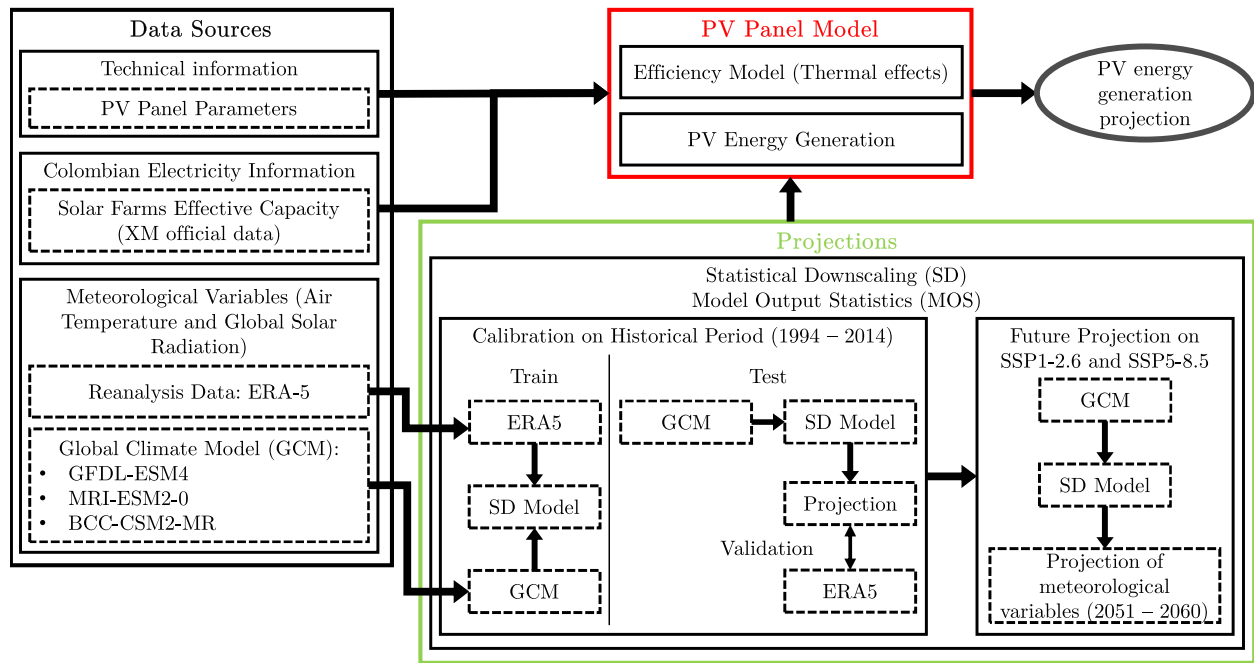


Fig. 1. Flowchart for future PV-energy generation calculation.

Agustín Codazzi (IGAC), which provides the standard geographic and climatic zoning used for planning and environmental assessments in Colombia. This classification is widely adopted in national studies and ensures consistency in regional-scale analyses. In this study, the eastern departments within the Orinoquía region and the Amazon region are not considered, as these areas have very low population density and currently lack significant solar infrastructure or concrete plans for large-scale photovoltaic deployment, making them less relevant for the present assessment of climate change impacts on PV energy generation.

Fig. A.14(a) shows the location of Colombia (red) in the American continent and Fig. A.14(b) shows the departments included in the study area (yellow) in Colombia. In addition, a map illustrating the division of the study area into the climatic regions considered in this study is provided to support the spatial aggregation framework used throughout the analysis (Fig. A.14(c)).

2.2. Data sources

ERA5 reanalysis data are used as the reference dataset for the calibration and evaluation of CMIP6 outputs. Produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) [32,33], ERA5 combines a wide range of global observations with numerical weather prediction models to provide consistent, high-resolution meteorological fields. In this study, 3-hourly surface downwelling shortwave radiation and 2 m air temperature are extracted from ERA5, as these variables are required for the subsequent modeling framework. ERA5 is widely adopted as a benchmark for bias correction and data quality assessment in climate and energy applications [20,34,35], and has been shown to outperform other reanalysis products such as MERRA-2 and JRA-55 in representing solar radiation [36,37].

Future and historical climate projections are obtained from the CMIP6, coordinated by the Working Group on Coupled Modelling under the World Climate Research Programme. CMIP6 provides standardized outputs from global climate models for both historical conditions and future climate scenarios driven by Shared Socioeconomic Pathways (SSPs). This study uses 3-hourly simulations of surface downwelling shortwave radiation and near-surface air temperature from the historical period 1994–2014 and from two future scenarios covering 2051–2060: SSP1–2.6 and SSP5–8.5. The selected general circulation

models are GFDL-ESM4 [38], MRI-ESM2-0 [39], and BCC-CSM2-MR [40], which were chosen due to their relatively fine spatial resolution and because they are among the few CMIP6 models providing meteorological outputs at a 3-hour temporal resolution. These scenarios represent low and high greenhouse gas emission pathways, respectively, with SSP5–8.5 projecting a global mean temperature increase of approximately 3.3 to 5.7 °C by the end of the century relative to pre-industrial levels.

Official information on solar farms was obtained from XM, the operator of the Colombian electricity market, through its open application programming interface (API) [26]. This dataset provides up-to-date records of the installed capacity of photovoltaic installations currently integrated or about to begin operations, and constitutes the basis for representing the spatial distribution of photovoltaic generation across the country (Fig. A.16). The installed capacity data was combined with downscaled meteorological projections and photovoltaic panel model to estimate future energy generation under different climate change scenarios.

Finally, global horizontal irradiation and 2-m air temperature data were retrieved from IDEAM (Instituto de Hidrología, Meteorología y Estudios Ambientales) [41] to validate statistical downscaling models against the following stations: Universidad Tecnológica [15015120], Paici [15075150], Universidad Nacional [21205012], Tunguavita [24035430], Puerta Roja [25025350], Armenia [26125290], Emas [26155230], Fedearroz [28035060], La Palomera [35185010], Botana [52055210], Aeropuerto Guapi [53045040].

2.3. Climate change projections

Future climate projections are incorporated using statistical downscaling (SD) techniques within a model output statistics (MOS) framework, which adjust or correct bias of global climate model outputs based on historical observations to enhance their reliability at the regional scale. Bias correction refers to a set of statistical techniques designed to remove systematic deviations between climate model outputs and observations or reanalysis data. These methods adjust the distribution of simulated variables so that their statistical properties (e.g., mean, variance, and quantiles) are more consistent with the reference dataset, while preserving the temporal evolution projected

Table 1

Statistical downscaling models implemented for each meteorological variable. The suffix “-W” indicates the use of moving windows for the corresponding model.

Methodology	Reference	Model label	Meteorological variable	
			air temperature	Solar radiation
Additive Scaling	[45]	AS-W	•	
Empirical Quantile Mapping	[46]	EQM	•	•
		EQM-W	•	
Parametric Quantile Mapping	Gaussian: [42] Gamma: [47] Weibull: [48]	PQM	•	•
		PQM-gamma		•
		PQM-weibull		•
		VAR	•	•
Variance Scaling	[49]	VAR	•	•
Quantile Delta Mapping	[50]	QDM	•	•
Power Transformation	[51]	PTR		•

Table 2

Metrics used for statistical downscaling validation for each meteorological variable.

Metric	Meteorological variable	
	Air temperature	Solar radiation
Root Mean Square Error (RMSE)	•	•
Mean bias	•	•
Ratio of standard deviations	•	•
Skewness bias	•	•
98th percentile (P98) bias	•	•
2nd percentile (P02) bias	•	•

by the climate models. In this study, SD methods are applied to solar radiation and near-surface air temperature in order to characterize their future evolution under climate change conditions in Colombia and to quantify the resulting impacts on PV energy generation. These approaches rely on statistical relationships between large-scale predictors simulated by GCMs and locally observed meteorological variables over the study domain [30]. The downscaling procedure begins with a calibration phase, in which SD models are trained using paired historical GCM simulations and reference observations. The calibrated models are subsequently applied to future GCM projections to generate localized climate scenarios under different SSP pathways.

Among the available SD approaches, MOS methods are widely used due to their simplicity and effectiveness in correcting systematic biases in climate model outputs [42]. MOS techniques typically employ the GCM-simulated value of the target variable at the nearest grid point as the main predictor, allowing for direct bias adjustment relative to observations. However, the performance of MOS methods depends on both the meteorological variable and the regional climatic characteristics, making it necessary to evaluate multiple SD formulations. The SD methods evaluated in this study are summarized in Table 1. Some of these methods incorporate a moving-window approach in which bias corrections are estimated for specific periods of the year by centering the calibration on each day of the year (day). This strategy enables the derivation of seasonally varying correction functions rather than a single annual adjustment, thereby improving the representation of intra-annual variability [43]. The implementation of the SD framework and the required data transformations were performed using R packages developed by [44]. The performance of the model is assessed using the evaluation metrics listed in Table 2, which quantify the ability of each SD method to reproduce the observed distribution of meteorological variables.

2.3.1. Statistical downscaling validation

A validation was performed for each of the three GCMs (GFDL-ESM4, MRI-ESM2-0, and BCC-CSM2-MR) and for both meteorological variables included in the analysis. Model performance was evaluated by comparing downscaled results against observed reanalysis data using multiple statistical metrics. Historical ERA5 and CMIP6 data spanning 1994–2014 were divided into calibration and validation subsets. The period 1994–2007 was used to calibrate the statistical downscaling

models, whereas the independent period 2008–2014 was employed to evaluate model performance and support the selection of the most suitable bias-correction approach.

The validation results for the statistical downscaling (SD) models applied to air temperature are summarized in Fig. 2. The root mean square error (RMSE), shown in Fig. 2(a), indicates that, for the BCC-CSM2-MR and MRI-ESM2-0 GCMs, the QDM method yields the lowest values, closely followed by EQM and the VAR approach. In contrast, for GFDL-ESM4, the PQM method performs best according to this metric. The mean bias results (Fig. 2(b)) indicate that most SD methods produce near-zero biases across all GCMs, with QDM consistently achieving the smallest deviations, followed by EQM and VAR. Analysis of the standard deviation ratio (Fig. 2(c)) reveals that the AS-W, EQM-W, and PQM methods systematically fail to reproduce the observed variability, whereas EQM, VAR, and QDM better preserve the distributional spread of air temperature for all GCMs. Further evaluation using the skewness bias metric (Fig. 2(d)) highlights limitations of the VAR method, which exhibits notable deviations, while EQM and QDM accurately capture the asymmetry of the observed temperature distributions across all models. Extreme temperature behavior is assessed using the P98 and P02 bias metrics, representing high- and low-temperature events, respectively. As shown in Fig. 2(e), QDM and EQM exhibit the lowest biases for extreme high temperatures, with particularly robust performance for BCC-CSM2-MR and GFDL-ESM4, where boxplot values remain close to zero. Consistent results are observed for low-temperature extremes in Fig. 2(f), where QDM demonstrates superior stability and accuracy, especially for GFDL-ESM4. Based on its overall performance across central tendency, variability, distributional shape, and extremes, QDM is selected as the most suitable SD method for downscaling air temperature over the study area. All future projections presented in the Results section are based on statistically downscaled CMIP6 climate projections using the QDM method. The results are reported for the mid-century period (2051–2060) relative to the historical baseline period (1994–2014) under the SSP1–2.6 and SSP5–8.5 emission scenarios. Unless otherwise specified, only statistically significant changes at the 95% confidence level are displayed in the figures. The percentage of national area exhibiting statistically significant changes is shown in the upper-left corner of each panel.

For solar radiation, the validation results of the SD models are summarized in Fig. A.15. In Fig. A.15(a), RMSE indicates that the PQM method consistently performs poorly across all GCMs. In contrast, the remaining SD approaches exhibit comparable RMSE values, including PQM-gamma and PQM-weibull, which display similar medians and interquartile ranges in their boxplots. The mean bias analysis (Fig. A.15(b)) further confirms the weak performance of PQM for all three GCMs, while the other SD methods show broadly similar and near-zero biases. According to this metric, MRI-ESM2-0 exhibits systematically larger biases than the other GCMs. The standard deviation ratio results (Fig. A.15(c)) reveal that, in addition to PQM, the PQM-gamma and PQM-weibull variants also struggle to reproduce the observed variability of solar radiation. By contrast, EQM, PTR, VAR, and QDM demonstrate a close agreement with the observed standard deviation

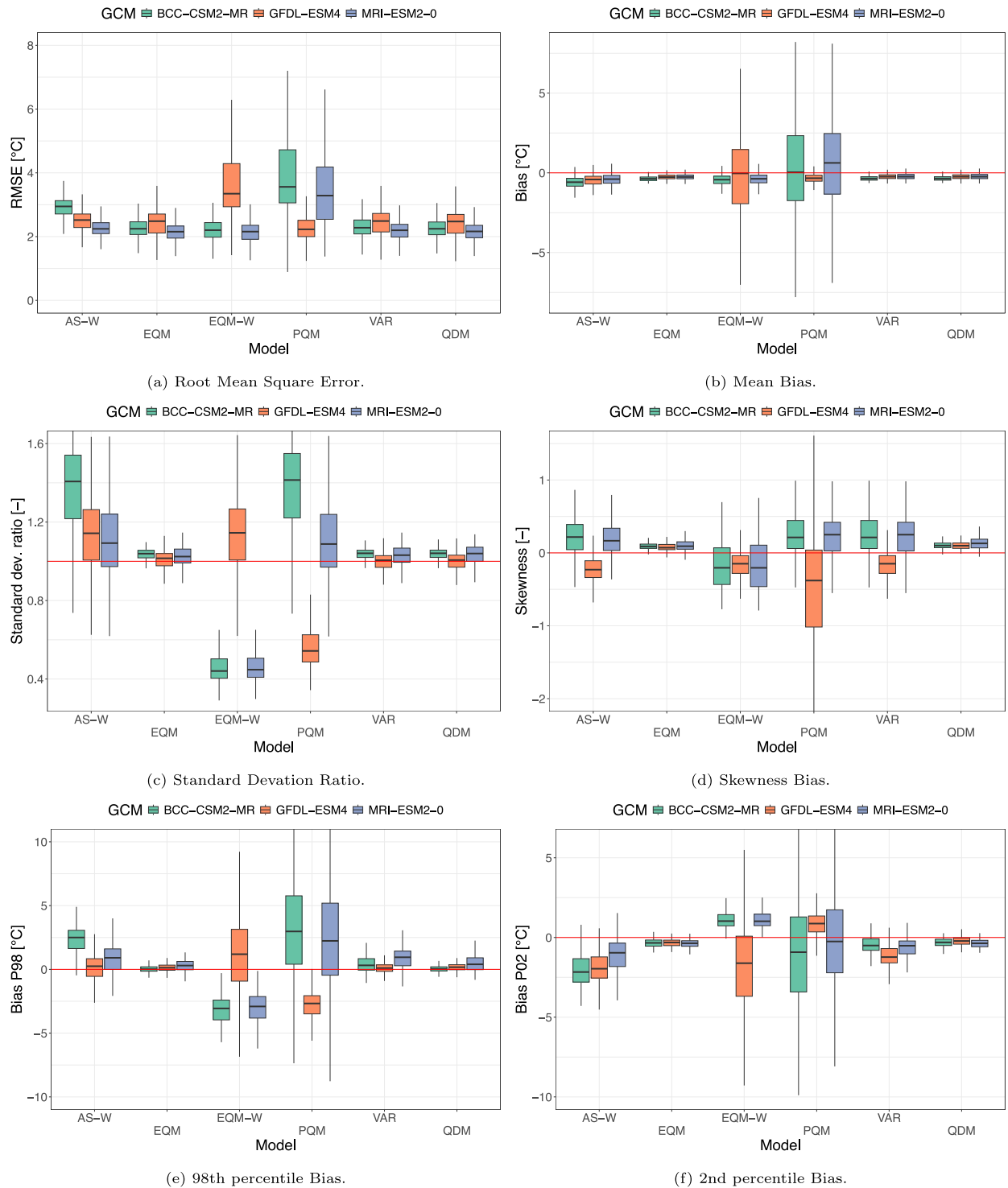


Fig. 2. Validation metrics for air temperature of the SD models for the three GCMs. Each boxplot represents the performance of an individual model for a given variable and metric. Missing points indicate cases where the SD model exhibited poor performance and was therefore excluded from further analysis. Abbreviations: AS-W (Additive Scaling), EQM and EQM (Empirical Quantile Mapping), PQM (Parametric Quantile Mapping), VAR (Variance Scaling), QDM (Quantile Delta Mapping), PTR (Power Transformation).

across all GCMs. Analysis of the skewness bias (Fig. A.15(d)) highlights further limitations of the PQM-based methods, as well as PTR and VAR, which exhibit notable departures from the observed distributional asymmetry. In comparison, EQM and QDM consistently capture the skewness of solar radiation distributions for all evaluated GCMs.

Extreme solar radiation behavior is evaluated using the P98 and P02 bias metrics, representing high- and low-radiation events, respectively.

As shown in Fig. A.15(e), EQM and QDM achieve the lowest biases for both extremes, with particularly robust performance for BCC-CSM2-MR and GFDL-ESM4 in estimating high-radiation events, and for GFDL-ESM4 in capturing low-radiation extremes, where boxplot medians remain close to zero. Considering its balanced performance in reproducing the mean state, variability, distributional shape, and extremes, EQM is selected as the most suitable SD method for downscaling solar

Table 3

Statistical validation of the downscaled temperature against observations from IDEAM meteorological stations using independent two-sample t-tests.

Station	Universidad Tecnológica	Paici	Universidad Nacional	Tunguavita	Puerta Roja	Armenia
p-value	0.3160	0.1640	0.5585	0.3636	0.5574	0.3980
Station	Emas	Fedearroz	La Palomera	Botana	Aeropuerto Guapi	
p-value	0.6714	0.0723	0.1968	0.0846	0.1190	

Table 4

Statistical validation of the downscaled solar radiation against observations from IDEAM meteorological stations using independent two-sample t-tests.

Station	Universidad Tecnológica	Paici	Universidad Nacional	Tunguavita	Puerta Roja	Armenia
p-value	0.1306	0.3619	0.2037	0.0916	0.1845	0.0982
Station	Emas	Fedearroz	La Palomera	Botana	Aeropuerto Guapi	
p-value	0.1882	0.6071	0.3586	0.4148	0.1936	

radiation over the study area. All future projections presented in the Results section are based on statistically downscaled CMIP6 climate projections using the EQM method. The results are reported for the mid-century period (2051–2060) relative to the historical baseline period (1994–2014) under the SSP1–2.6 and SSP5–8.5 emission scenarios. Unless otherwise specified, only statistically significant changes at the 95% confidence level are displayed in the figures. The percentage of national area exhibiting statistically significant changes is shown in the upper-left corner of each panel.

2.3.2. Model weighting assessment

Uncertainty in climate projections is commonly assessed using ensembles of global climate models, where each model is often treated as an independent realization of future climate conditions. However, recent studies have shown that CMIP6 models exhibit substantial structural interdependence due to shared components and development lineages, which can lead to an overestimation of confidence when applying equal weighting schemes [52,53]. To address this limitation, we apply a genealogy-aware model weighting approach following [53], which accounts for both model performance and interdependence within the ensemble. This framework introduces two key dispersion parameters, σ_D and σ_S , which control the influence of model diversity and similarity in the weighting procedure. The resulting weights reduce redundancy among structurally related models and provide a more robust characterization of projection uncertainty.

For solar radiation, the estimated parameters are $\sigma_D = 0.15$ and $\sigma_S = 0.55$, resulting in weights of 0.2962 for GFDL-ESM4, 0.3353 for MRI-ESM2-0, and 0.3685 for BCC-CSM2-MR. For air temperature, the calibrated parameters are $\sigma_D = 0.2$ and $\sigma_S = 0.45$, yielding weights of 0.2267, 0.4368, and 0.3363, respectively. These weights are used to compute the ensemble projections presented in this study.

2.3.3. Validation against in situ observations

Although ERA5 is widely used as a reference dataset for bias correction and statistical downscaling applications [20,34,35], reanalysis products are not direct observations and may contain uncertainties associated with model physics, data assimilation procedures, and observational coverage. This limitation is particularly relevant in regions with complex topography, such as the Colombian Andes, where local atmospheric processes may not be fully resolved. To provide an independent evaluation of the proposed methodology, the downscaled solar radiation and air temperature series were validated against observations from eleven IDEAM meteorological stations distributed across Colombia. Independent two-sample t-tests were performed to assess whether statistically significant differences existed between modeled and observed mean values for both variables. Tables 3 and 4 summarize the resulting p-values for air temperature and solar radiation, respectively.

For all stations analyzed, the null hypothesis of equal means could not be rejected at the 95% confidence level. Consequently, no statistically significant differences were identified between the downscaled

and observed series for either solar radiation or air temperature. These results provide independent observational evidence that the proposed statistical downscaling framework adequately reproduces local climatic conditions relevant for photovoltaic energy assessment. Nevertheless, the use of ERA5 as the primary reference dataset remains a source of uncertainty that should be considered when interpreting future climate projections.

2.4. Photovoltaic power generation

The power generation potential of a photovoltaic (PV) system is governed by the interaction between incident solar irradiance and module efficiency, which depends on irradiance intensity, cell temperature, and intrinsic panel characteristics. The PV power output at a given location can be estimated by multiplying the installed capacity by the PV power potential (PVP_{pot}), a dimensionless metric representing the performance of PV modules under prevailing environmental conditions [54].

Two main approaches are commonly used to estimate PVP_{pot} : the power rating method and the energy rating method. The power rating method integrates instantaneous PV power output over time, whereas the energy rating method estimates PV potential by multiplying the accumulated solar irradiance over a given period by a performance ratio [55]. The latter approach has been widely adopted in recent studies [54,55] and is also applied in this work. Accordingly, PV power potential is calculated as:

$$PVP_{pot}[t] = P_r[t] \frac{I_s[t]}{I_{STC}} \quad (1)$$

where $I_s[t]$ denotes surface solar irradiance, I_{STC} is the irradiance under Standard Test Conditions (1000 W/m^2), and $P_r[t]$ is the performance ratio, which accounts for temperature-induced efficiency variations. The performance ratio is defined as:

$$P_r[t] = 1 + \beta \cdot (T_{cell}[t] - T_{STC}) \quad (2)$$

where T_{STC} is the reference temperature under STC (25°C), and β is the temperature coefficient specific to the PV technology. In this study, a mono-crystalline silicon module is assumed, with $\beta = -0.005^\circ\text{C}^{-1}$.

The cell temperature T_{cell} can be estimated using different parameterizations that incorporate solar irradiance, air temperature, wind speed, and relative humidity. Although some formulations explicitly account for wind-induced convective cooling, previous studies suggest that wind speed and relative humidity exert a limited influence in certain climatic conditions. Therefore, we adopt the formulation used in previous studies [55]:

$$T_{cell}[t] = c_1 + c_2 \cdot T_a[t] + c_3 \cdot I_s[t] \quad (3)$$

where $T_a[t]$ represents ambient air temperature. The coefficients for mono-crystalline silicon technology are $c_1 = 3.75^\circ\text{C}$, $c_2 = 1.14$, and $c_3 = 0.0175^\circ\text{C m}^2 \text{ W}^{-1}$.

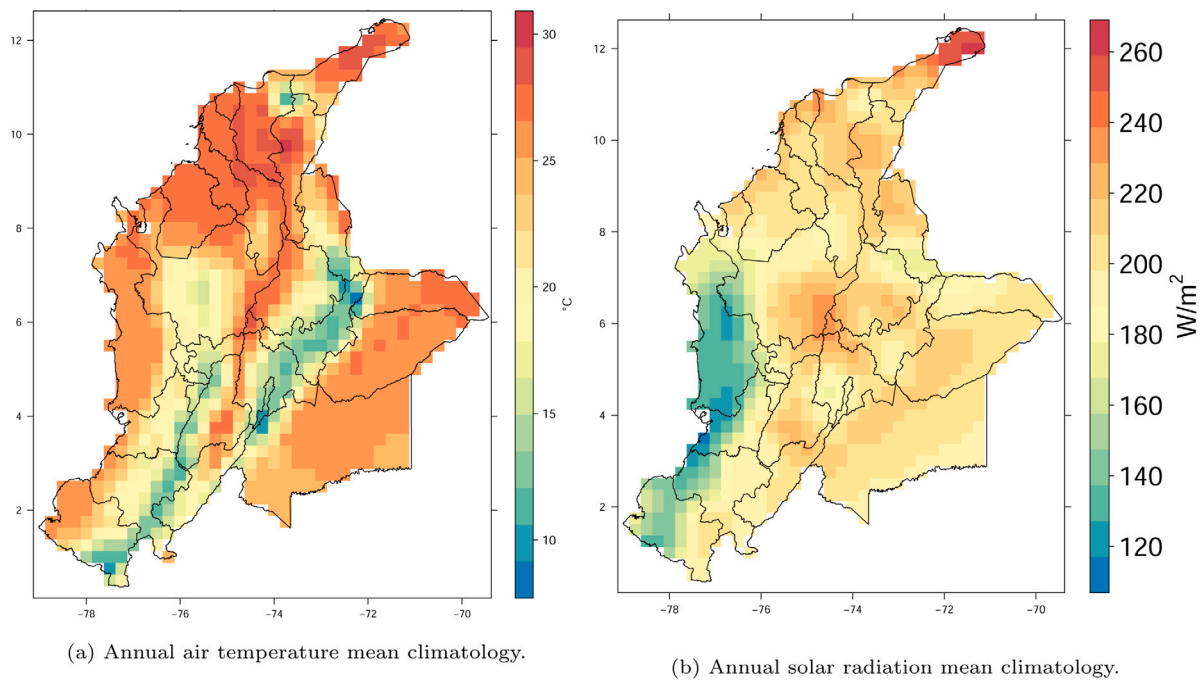


Fig. 3. Mean solar radiation (W/m^2) (a) and air temperature ($^{\circ}\text{C}$) (b) over study area in Colombia during the historical period (1994–2014).

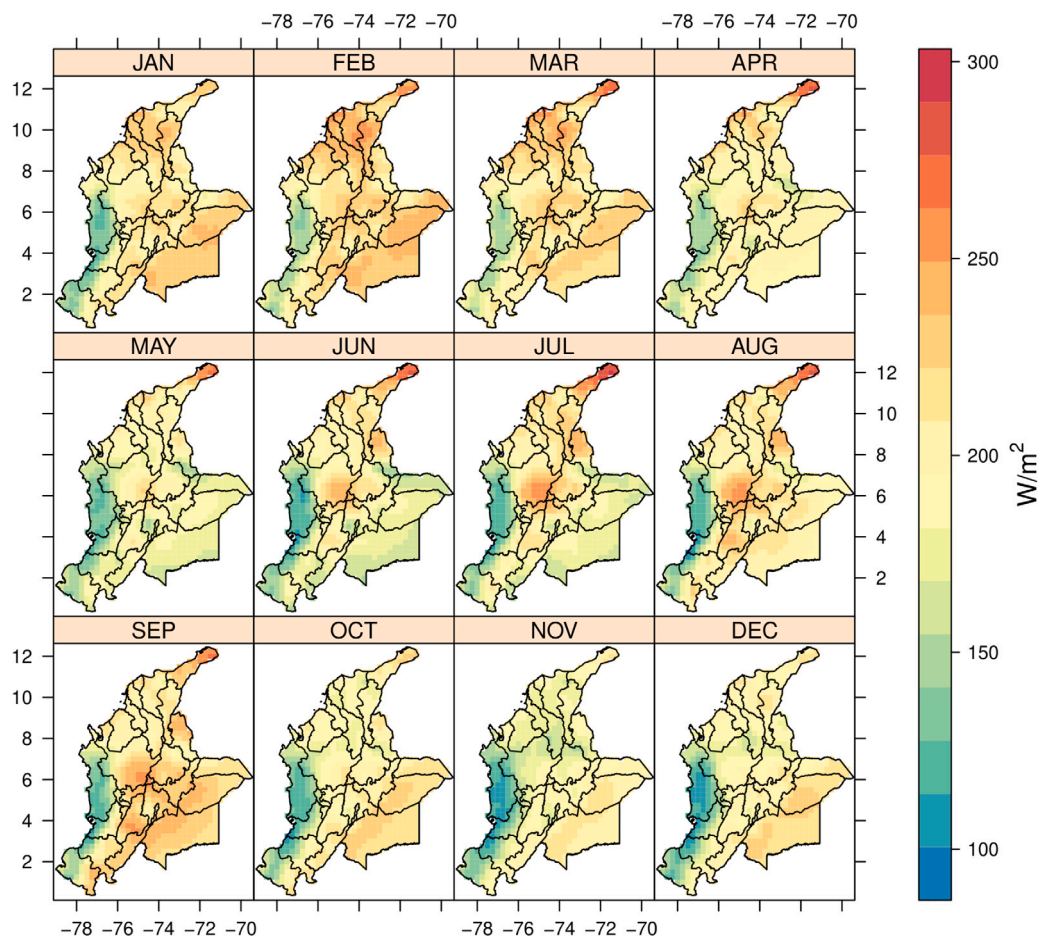


Fig. 4. Monthly variability in solar radiation over study area in Colombia during the historical period (1994–2014).

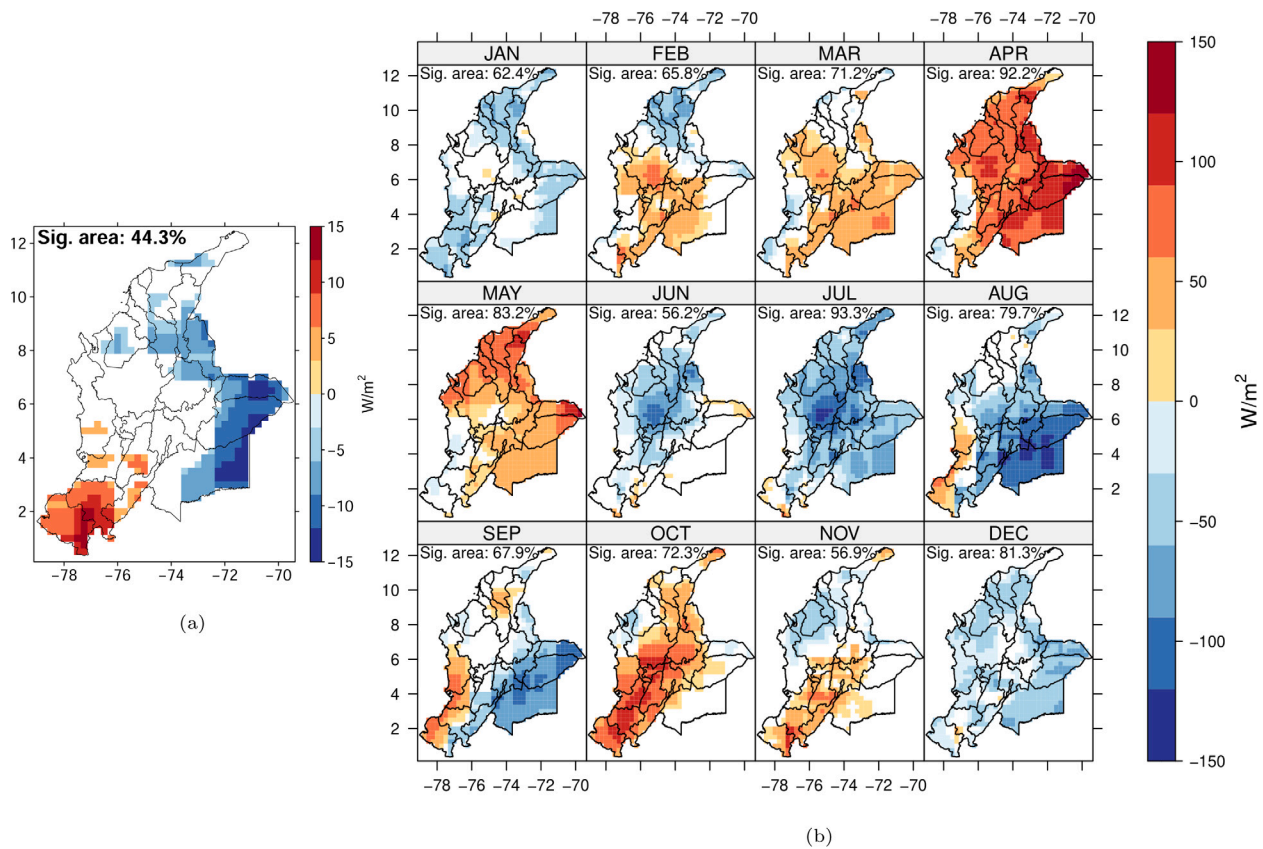


Fig. 5. Projected changes in surface solar radiation under SSP5-8.5. (a) Annual mean change (W/m^2). (b) Monthly mean change (W/m^2). See Section 2.3 for details.

3. Results

3.1. Climatology of meteorological variables

Figs. 3(a) and 3(b) present the mean annual climatology values for air temperature and solar radiation (respectively), while Fig. 4 presents the mean monthly climatology values for solar radiation. Near surface air temperature in Colombia is strongly controlled by topography, reflecting the dominant influence of altitude across the country (Fig. 3(a)). The lowest mean annual temperatures are observed over the Andean region, where elevations exceeding 2000 m above sea level lead to substantially cooler conditions, with a mean value of 19.09°C . In contrast, lowland regions exhibit significantly higher temperatures, particularly across the Eastern Llanos, and the Caribbean and Pacific coasts, where mean annual air temperatures of 25.7 , 26.3 and 25.5°C , respectively. These warmer conditions are associated with low elevations and limited thermal variability throughout the year. Overall, the pronounced spatial gradient in air temperature highlights the strong coupling between Colombia's complex topography and its thermal climate.

In Fig. 3(b), the maximum annual mean solar radiation values ($>250 \text{ W/m}^2$) are observed over the Guajira Peninsula in northern Colombia, a semi-arid to desert region characterized by low cloud cover and limited solar radiation, which together favor high solar energy potential. In contrast, the lowest annual mean values ($<190 \text{ W/m}^2$) are found in the Pacific Coast, where persistent cloudiness and some of the highest solar radiation rates in the world substantially reduce surface solar irradiance. Across the remaining portions of the study domain, annual mean solar radiation values are generally close to 200 W/m^2 , reflecting the combined influence of complex topography, frequent cloud formation, and regional moisture transport.

Seasonal variability in solar radiation is closely linked to Colombia's bimodal solar radiation regime, which is driven by the annual migration of the Intertropical Convergence Zone (ITCZ). Periods of enhanced rainfall and cloud cover during April–May and October–December are associated with pronounced reductions in incoming solar radiation, particularly over the Andean Region and Pacific Coast (Fig. 4). Conversely, relatively higher solar radiation levels are observed during the drier seasons, especially from February to March and during July–August in northern and central part of Colombia, when cloud cover is reduced.

3.2. Climate change projections

3.2.1. Solar radiation and air temperature

Fig. 5 summarizes the projected changes in surface solar radiation for the mid-century period (2051–2060) relative to the historical baseline (1994–2014) under the high-emission SSP5–8.5 scenario. Fig. 5(a) presents the spatial distribution of annual mean changes, highlighting a pronounced regional contrast between solar radiation increases (brightening) and decreases (dimming) across the study area. The strongest annual dimming is projected over the Eastern Llanos, where mean solar radiation is expected to decrease by approximately 10 – 15 W/m^2 . A more moderate but still substantial reduction is projected for the northern Caribbean Coast, which currently hosts a large share of Colombia's installed solar capacity, with decreases ranging from 3 to 8 W/m^2 . In contrast, the southwestern Pacific Coast exhibits a general brightening signal, with projected annual mean increases of approximately 7.5 – 13.5 W/m^2 . This spatial pattern suggests an emerging contrast in which regions with historically high photovoltaic potential tend to experience future dimming, whereas areas with lower baseline potential show relative increases in solar radiation, likely associated with projected changes in cloud cover and precipitation regimes.

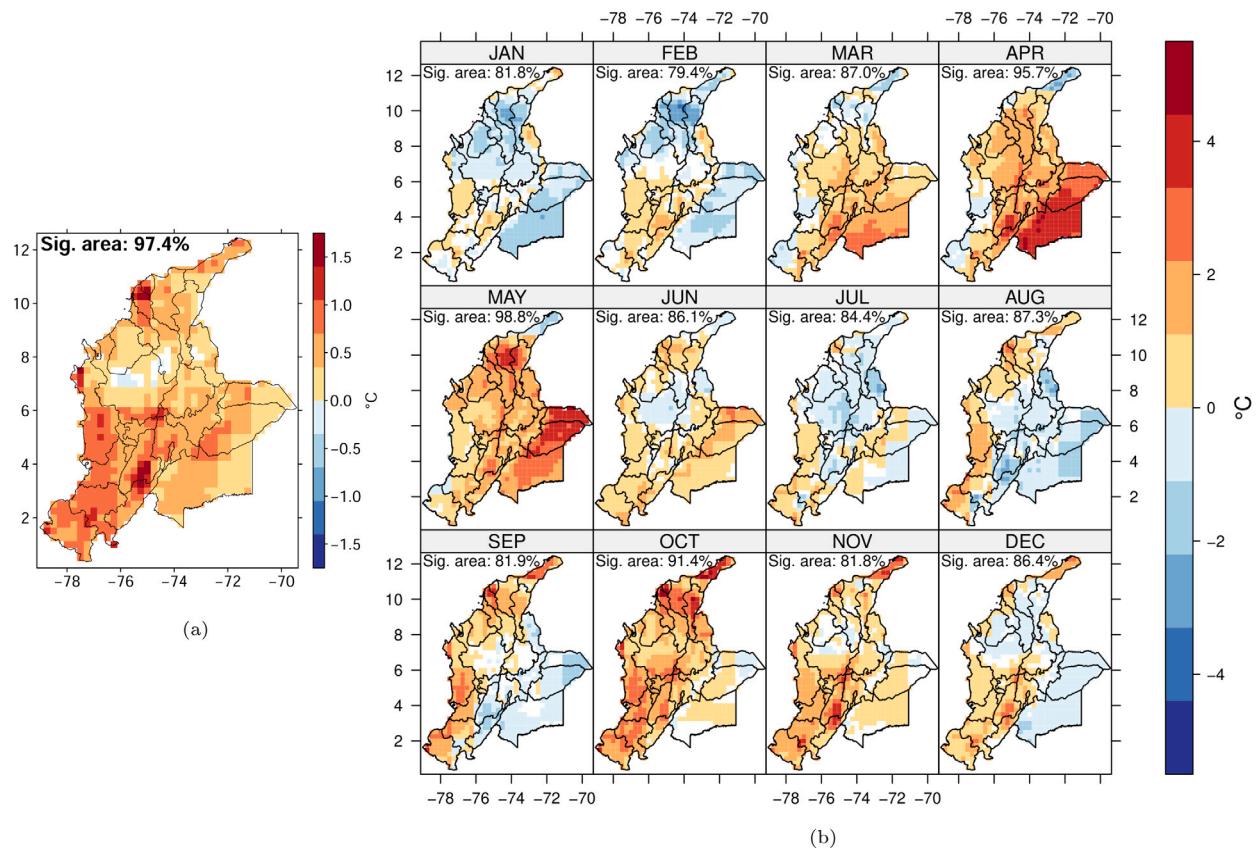


Fig. 6. Projected changes in air temperature under SSP5-8.5. (a) Annual mean change (°C). (b) Monthly mean change (°C). See Section 2.3 for details.

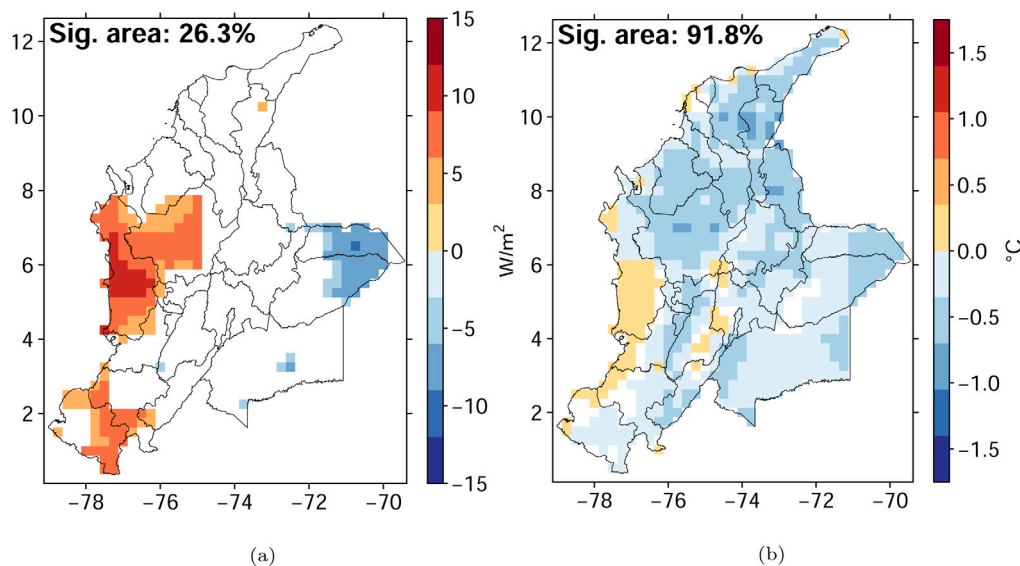


Fig. 7. Projected changes in the annual mean of solar radiation (W/m²) (a) and air temperature (°C) (b) under SSP1-2.6 scenario. See Section 2.3 for details.

The intra-annual evolution of projected solar radiation changes is illustrated in Fig. 5(b), which presents monthly anomalies. The projections reveal a pronounced seasonal structure closely linked to Colombia's rainfall regime. Significant brightening is projected over more than 70% of the study area during April, May, and October, with mean monthly increases on the order of 70–140 W/m². In contrast, the strongest dimming occurs during July and August, when reductions of approximately 50–130 W/m² are projected in several regions. These results reflect the marked seasonality of Colombia's climate system, in

which projected changes in cloudiness and precipitation modulate solar radiation in a manner consistent with the country's well-defined wet and dry seasons.

Fig. 6 presents the projected changes in near-surface air temperature (during daylight hours) for the mid-century period (2051–2060) relative to the historical reference period (1994–2014) under the SSP5–8.5 scenario. Fig. 6(a) shows the spatial distribution of annual mean temperature anomalies, indicating a generalized warming signal throughout the study area. Projected annual mean temperature increases range

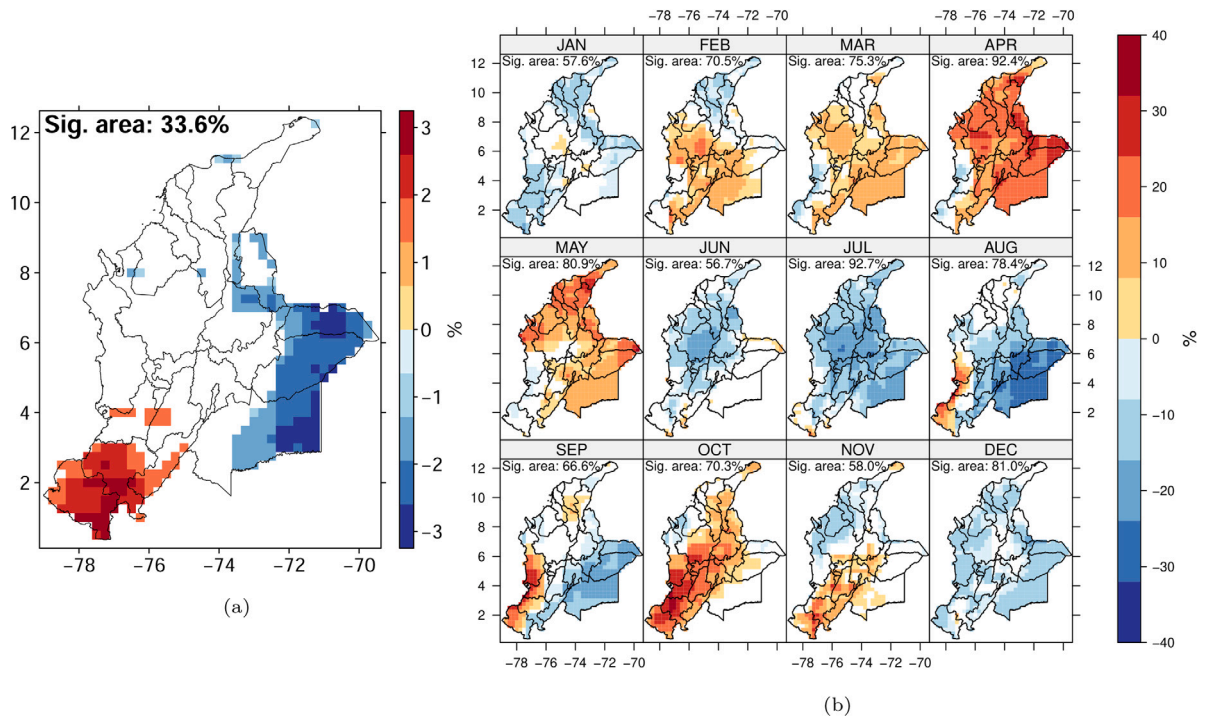


Fig. 8. Projected changes in PV power generation under SSP5-8.5. (a) Annual mean change (%). (b) Monthly mean change (%). See Section 2.3 for details.

from approximately $+0.25$ to $+1.65$ °C, with the largest warming concentrated in regions such as the central Caribbean coast, the Pacific region, and the central and southern Andean region. This widespread warming is consistent with expectations under a high-emission climate change pathway and is particularly relevant for photovoltaic systems, as higher operating temperatures are associated with reduced panel efficiency due to thermal effects. The intra-annual evolution of temperature changes is illustrated in Fig. 6(b), which shows monthly variations and reveals a clear seasonal modulation linked to regional rainfall patterns, similar to that observed for solar radiation. The magnitude of warming varies by month and region, with the Eastern Llanos region experiencing its strongest temperature increases between April and May, reaching monthly variations of up to approximately $+3$ – 5 °C, while the central Caribbean coast exhibits a more pronounced warming in May and October.

Fig. 7 summarizes the projected changes in solar radiation and daytime air temperature for the mid-century period (2051–2060) relative to the historical baseline (1994–2014) under the SSP1–2.6 scenario. Fig. 7(a) presents the spatial distribution of annual mean solar radiation changes, showing fewer regions with statistically significant variations compared to those projected under SSP5–8.5 across the study area. The most pronounced dimming is projected over the eastern region, with annual reductions of up to approximately 9 W/m², while the strongest brightening is confined to a limited area along the Pacific coast, reaching maximum increases of about 10 W/m². In contrast to SSP5–8.5, the Caribbean region does not exhibit significant dimming under SSP1–2.6, highlighting the sensitivity of regional solar radiation patterns to the emission pathway considered. For daytime air temperature (Fig. 7(b)), projections indicate a slight but statistically significant cooling signal across Colombia, with annual mean decreases remaining below 1 °C.

3.2.2. Photovoltaic power generation

Figs. 8 and 9 present the projected changes in photovoltaic power generation for the mid-century period (2051–2060) relative to the historical baseline (1994–2014) under the SSP5–8.5 and SSP1–2.6 scenarios, respectively, showing both annual and monthly percentage changes. Under the high-emission SSP5–8.5 scenario (Fig. 8(a)), the

spatial and seasonal patterns of PV generation closely mirror those obtained for solar radiation, where the strongest annual reductions in PV generation are projected over the eastern Llanos region, with decreases ranging between about -2% and -3.15% . In contrast, the southwestern part of Pacific region shows an increase signal, where annual mean increases of roughly 2% to 3.15% are projected. The intra-annual evolution of PV generation under SSP5–8.5, shown in Fig. 8(b), further confirms this close association with solar radiation variability. During April, May, and October, PV generation is projected to increase across most of the study area, with mean monthly gains on the order of 25% – 35% . Conversely, the most pronounced reductions occur during July and August, when decreases of approximately -20% to -32% are simulated in several regions.

Under the low-emission SSP1–2.6 scenario (Fig. 9), the spatial distribution of annual changes in photovoltaic generation reveals statistically significant increases on the Caribbean coast, the central region and the southwestern sector of the Pacific coast, with projected gains ranging from approximately 1.5% to 2.75% . Unlike the high-emission scenario, these increases are not directly associated with annual mean changes in solar radiation. Instead, they likely reflect the influence of intra-annual variability and the comparatively weaker warming signal under SSP1–2.6, which reduces temperature-induced efficiency losses and thereby enhances future PV potential. To further examine this relationship, the isolated contribution of air temperature to the projected changes in photovoltaic generation was quantified by keeping solar radiation constant under both SSP5–8.5 and SSP1–2.6 scenarios. Results indicate that, under SSP1–2.6, the temperature-driven contribution is not statistically significant across the study area. In contrast, under SSP5–8.5, the projected warming exerts a measurable negative impact in several regions, with annual mean reductions in PV generation ranging from approximately -0.75% to -1.6% .

Fig. 10 presents the projected impacts of climate change on photovoltaic (PV) energy production at currently operating solar farms in Colombia under the SSP1–2.6 and SSP5–8.5 scenarios. On an annual basis, the low-emission scenario indicates a moderate increase in PV generation ($+1.21\%$, $p_{\text{value}} < 0.002$). In contrast, the high-emission SSP5–8.5 scenario projects a nearly negligible annual decrease

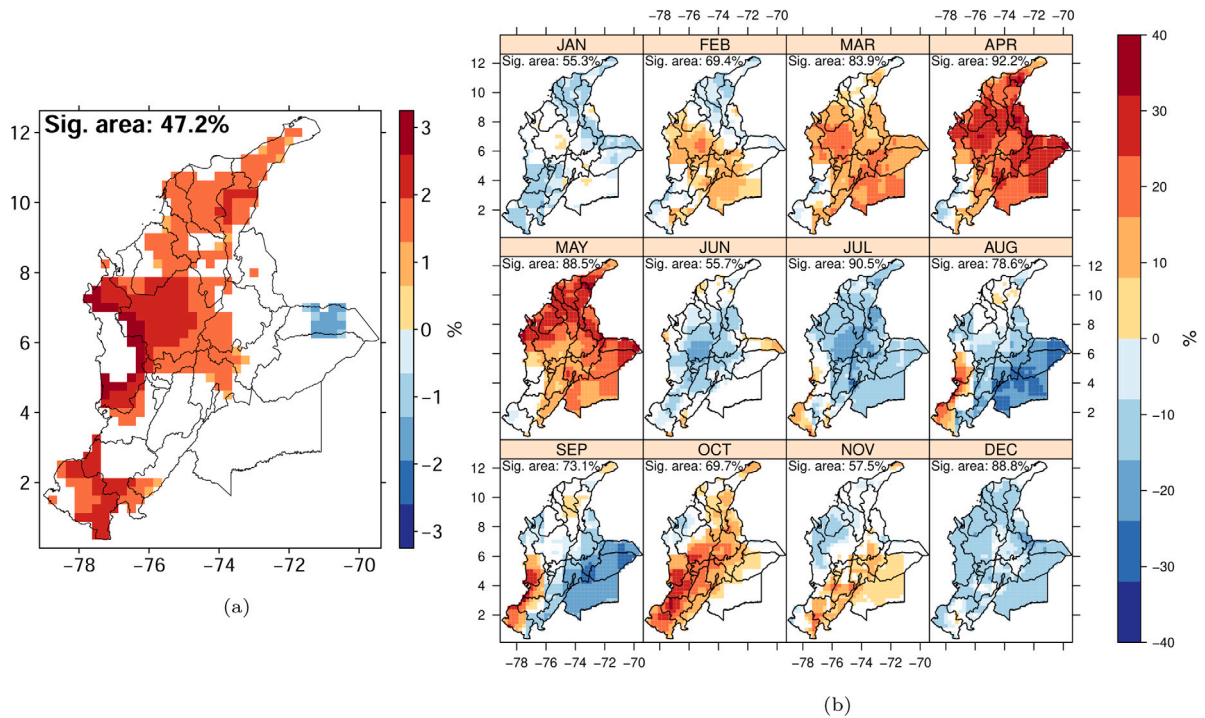


Fig. 9. Projected changes in PV power generation under SSP1-2.6. (a) Annual mean change (%). (b) Monthly mean change (%). See Section 2.3 for details.

(-0.37% , $p_{value} < 0.001$), suggesting that the negative effect of rising air temperatures is largely offset by annual mean increases in solar radiation. Nevertheless, monthly variations are considerably more relevant in the Colombian context due to their implications for seasonal complementarity within the national energy system. The results reveal a marked seasonal modulation of climate change impacts on PV generation, closely linked to rainfall-driven climatology of Colombia. Fig. 11 reveals that, under both scenarios, substantial increases in PV output are projected for April and May, where these enhancements coincide with projected brightening during transitional seasons. Conversely, the most pronounced and significant reductions occur in July, when PV generation declines under both emission pathways.

3.2.3. Sub-daily behavior under climate change

To further explore the impacts of climate change at sub-daily scales, Fig. 12 examines intra-day photovoltaic generation profiles derived from the 3-hourly projections. The figure compares historical conditions with future projections under SSP1-2.6 and SSP5-8.5 across four climatological trimesters (JFM, AMJ, JAS, and OND). The percentage change profiles (Figs. 12(a) and 12(b)), reveal a robust and consistent intra-day redistribution of photovoltaic generation under climate change. Across all trimesters and both scenarios, photovoltaic output tends to increase during the morning hours (07:00–09:00) while decreasing during the late afternoon (16:00–18:00). This pattern suggests a systematic shift in the solar resource availability rather than a uniform change in daily totals. Differences between scenarios are mainly observed during the morning period of JFM and AMJ, where the low-emission SSP1-2.6 scenario exhibits stronger increases, exceeding those of SSP5-8.5 by approximately 5 and 12 percentage points, respectively. These results highlight the importance of high-temporal-resolution climate data for capturing intra-day changes in photovoltaic performance that would remain unresolved at daily or monthly scales, and they are consistent with Colombia's marked seasonal and diurnal atmospheric dynamics.

It is important to note that Figs. 10, 11, and 12 provide complementary perspectives on the projected impacts of climate change on photovoltaic (PV) generation. Specifically, while Fig. 10 summarizes

the annual aggregated changes under future climate scenarios, Fig. 11 provides a more detailed seasonal perspective by highlighting statistically significant monthly variations and their underlying temporal structure. In addition, Fig. 12 complements both analyses by resolving sub-daily dynamics and capturing intra-day redistribution patterns in PV generation that are not observable at monthly or annual scales. Together, these figures offer a consistent multi-scale characterization of climate change impacts on PV resources, spanning annual, seasonal, and intra-day variability.

4. Discussion and conclusions

This study evaluated the impacts of climate change on PV power generation in Colombia using statistically downscaled CMIP6 projections, with an emphasis on seasonal and sub-daily variability enabled by 3-hourly data. A multi-model ensemble was constructed using the GCMs GFDL-ESM4, MRI-ESM2-0, and BCC-CSM2-MR, applying a genealogy-aware weighting scheme to account for both model performance and structural interdependence among CMIP6 models. This approach avoids the assumption of equal model independence and provides a more robust representation of projection uncertainty. The analysis focused on the combined effects of surface solar radiation and near-surface air temperature, the two primary meteorological drivers of PV system performance.

Projected changes in these variables reveal contrasting spatial behaviors. Air temperature exhibits a robust warming signal across the entire study area under SSP5-8.5. In contrast, changes in solar radiation are spatially heterogeneous, showing alternating patterns of brightening and dimming linked to projected modifications in cloud cover and precipitation regimes. As a result, future PV generation in Colombia is primarily driven by solar radiation variability, while temperature plays a secondary but systematic role through efficiency losses at higher operating temperatures.

At the annual scale, projected changes in PV generation are modest but spatially heterogeneous. Under the high-emission SSP5-8.5 scenario, the largest reductions are concentrated in regions with historically high photovoltaic potential, such as the Eastern Llanos and

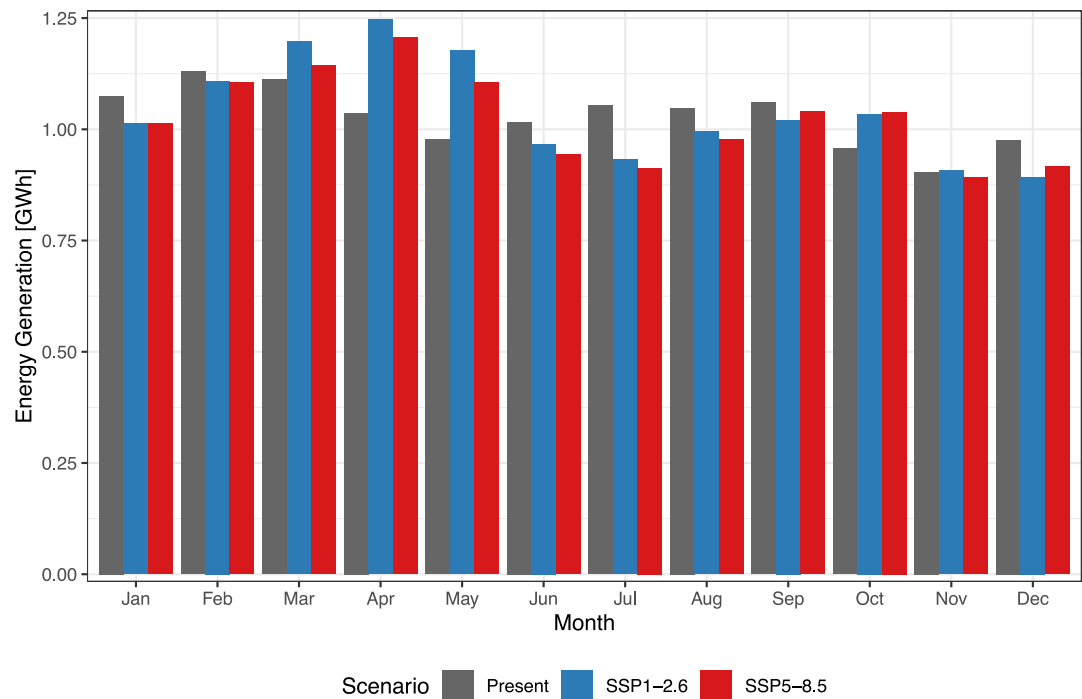


Fig. 10. Projected changes in photovoltaic energy generation for currently installed solar farms in Colombia under the SSP1-2.6 and SSP5-8.5 scenarios for the mid-century period (2051–2060) relative to the historical baseline (1994–2014).

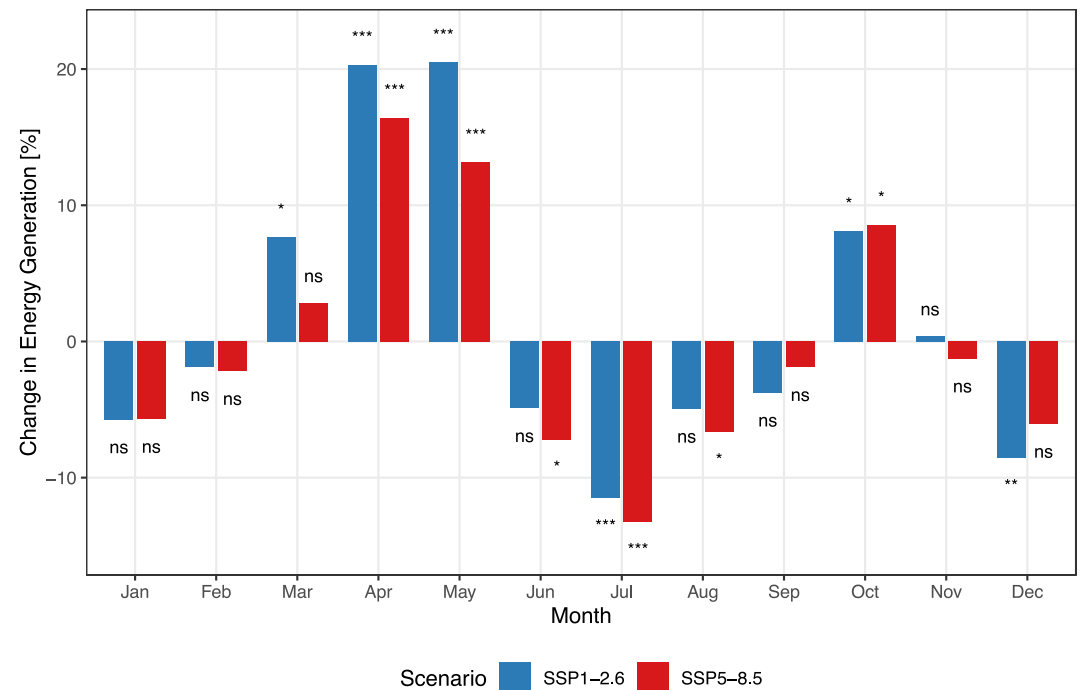


Fig. 11. Monthly photovoltaic energy generation changes under historical conditions and future climate scenarios (SSP1-2.6 and SSP5-8.5). Percentage changes are computed relative to the historical baseline. Statistical significance of monthly differences was assessed using two-sample t-tests comparing future and present distributions for each month. To account for multiple hypothesis testing across months, p-values were adjusted using the Benjamini–Hochberg false discovery rate (FDR) procedure. Significance levels are indicated as: $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***); non-significant results are labeled as “ns”.

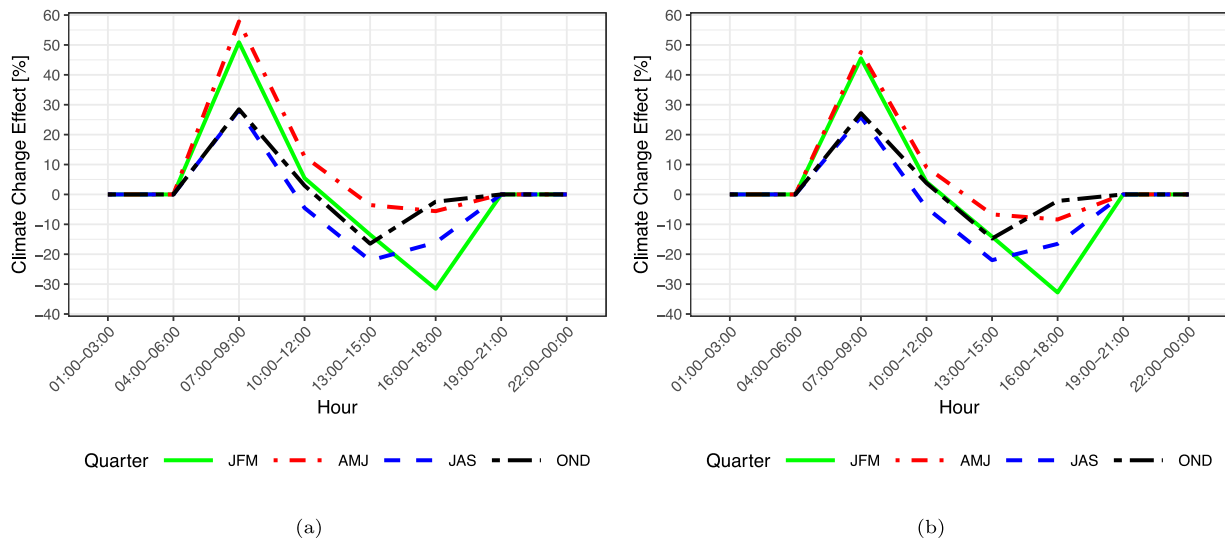


Fig. 12. Intra-day PV generation change profiles for Colombian solar farms derived from 3-hourly GCM data, shown for four trimesters (JFM, AMJ, JAS, OND). (a) Percentage changes under SSP1-2.6 relative to historical. (b) Percentage changes under SSP5-8.5 relative to historical.

parts of the Caribbean region, while increases are projected in central and southwestern Colombia. This spatial pattern suggests a partial redistribution of PV suitability, whereby areas with lower baseline potential experience relative gains. However, the annual signal masks more pronounced monthly contrasts that are far more relevant for power system operation and planning. At the seasonal scale, climate change modulates rather than fundamentally alters Colombia's solar climatology, but the magnitude of monthly PV anomalies is substantial. Projected increases during April, May, and October coincide with transitional rainfall periods, when hydropower availability may fluctuate, while reductions during mid-year months align with established dry-season dynamics in several regions. These results underscore that the added value of PV under climate change lies not in annual averages, but in its evolving seasonal complementarity with hydropower. Given the strong dependence of the Colombian electricity sector on hydropower, the projected monthly changes in PV generation may influence the temporal complementarity between renewable energy resources. Although such effects cannot be quantified within the scope of this study, the results suggest that seasonal variations in PV availability could be relevant for future assessments of system reliability, drought resilience, and dependence on thermal generation.

The use of 3-hourly climate projections allows for a detailed assessment of intra-day PV generation changes, revealing a consistent pattern across all trimesters: enhanced PV generation during morning hours (approximately 07:00–09:00) and reduced generation during late afternoon (16:00–18:00). This behavior persists throughout the year and reflects systematic atmospheric changes rather than isolated seasonal effects. Differences between scenarios are mainly confined to morning hours in JFM and AMJ, when the low-emission scenario exhibits larger positive anomalies. These projected sub-daily shifts may be relevant from a grid-integration perspective because they indicate changes not only in the magnitude but also in the timing of solar energy availability. In particular, enhanced morning generation and reduced afternoon production could affect the temporal distribution of PV output throughout the day. While the present study does not include storage, transmission, or demand modeling, these results provide useful information for future investigations focused on operational flexibility, storage deployment, and power-system planning under climate change conditions.

An additional aspect that may contribute to the differences observed between the SSP1-2.6 and SSP5-8.5 scenarios is the distinct evolution of aerosol precursor emissions embedded within the SSP narratives. Recent CMIP6 studies have shown that future photovoltaic potential is

influenced not only by greenhouse-gas-driven climate change but also by changes in atmospheric composition, particularly aerosol-radiation and aerosol-cloud interactions. Under low-emission pathways such as SSP1-2.6, reductions in aerosol loading can enhance the amount of solar radiation reaching the surface, potentially contributing to localized increases in photovoltaic potential. In contrast, higher-emission pathways involve more complex interactions among aerosols, clouds, and radiative processes, which may partially offset or amplify changes in solar radiation depending on regional atmospheric conditions.

Although the present study does not explicitly separate the contributions of aerosol forcing, cloud feedbacks, and greenhouse-gas-induced climate change, the contrasting responses obtained under SSP1-2.6 and SSP5-8.5 are consistent with the broader CMIP6 literature. Previous studies have reported that the beneficial effects associated with reduced aerosol emissions tend to be more evident under low-emission scenarios, whereas under high-emission pathways these effects may be overwhelmed by larger-scale atmospheric and radiative changes. Consequently, the projected changes in photovoltaic generation presented here should be interpreted as the combined outcome of multiple climate-system processes represented within each SSP scenario, rather than solely as a response to temperature and precipitation-driven changes in cloudiness.

This study provides a useful reference for the strategic and efficient deployment of solar photovoltaic energy in Colombia under a changing climate. Its relevance is underscored by the rapid global expansion of PV technologies, whose costs declined by more than 80% between 2010 and 2019, positioning solar energy as one of the main contributors to future electricity generation worldwide. Additionally, the results presented here are based on current crystalline silicon PV technologies with fixed efficiency parameters. However, ongoing technological advances—such as improvements in module efficiency, the development of thin-film and perovskite-based technologies, and enhanced thermal performance—may partially offset climate-induced efficiency losses in future systems. In parallel, progress in energy storage solutions, including batteries and green hydrogen, offers promising pathways to manage the inherent variability of solar generation and enhance system flexibility.

Taken together, these findings emphasize that climate change does not simply threaten or benefit PV generation in aggregate terms; rather, it reshapes its temporal distribution. Understanding this redistribution—across regions, seasons, and hours of the day—is essential for maximizing complementarity within Colombia's predominantly hydro-based electricity system. Given Colombia's strong reliance on

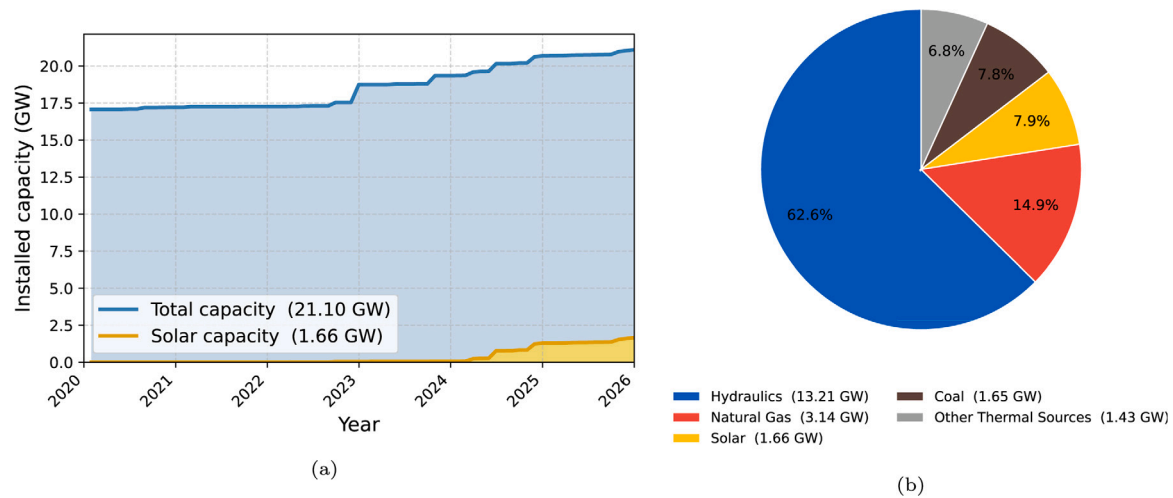


Fig. A.13. (a) Evolution of electricity installed capacity in Colombia over the period 2020–2025, showing the rapid expansion of solar photovoltaic generation alongside other technologies. (b) Electricity generation mix in 2025, presented as a percentage share of total installed capacity by source [26].

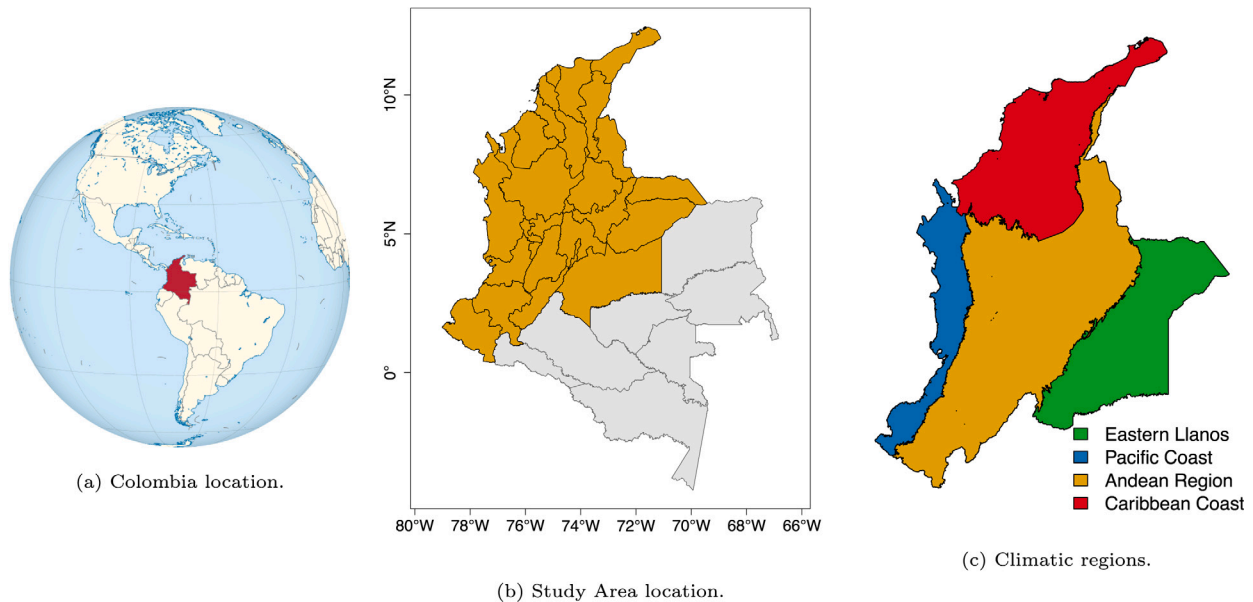


Fig. A.14. Geographical location of Colombia (a), the study area (b) and its subdivision in climatic regions (c).

hydropower and its sensitivity to climate variability, the projected changes in PV generation identified in this study may have implications for the future interaction between renewable energy resources. However, evaluating the extent of these interactions requires dedicated energy-system analyses that are beyond the scope of the present work. Accordingly, climate-informed planning, appropriate policy support, and targeted investments will be key to guiding the expansion of Colombia's solar energy sector.

CRediT authorship contribution statement

Carlos G. Ramírez-Ruiz: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sergio Cabrales:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Carlos F. Valencia:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to make the text more readable. After using this tool, the authors reviewed and edited the content as needed and assume full responsibility for the content of the publication.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

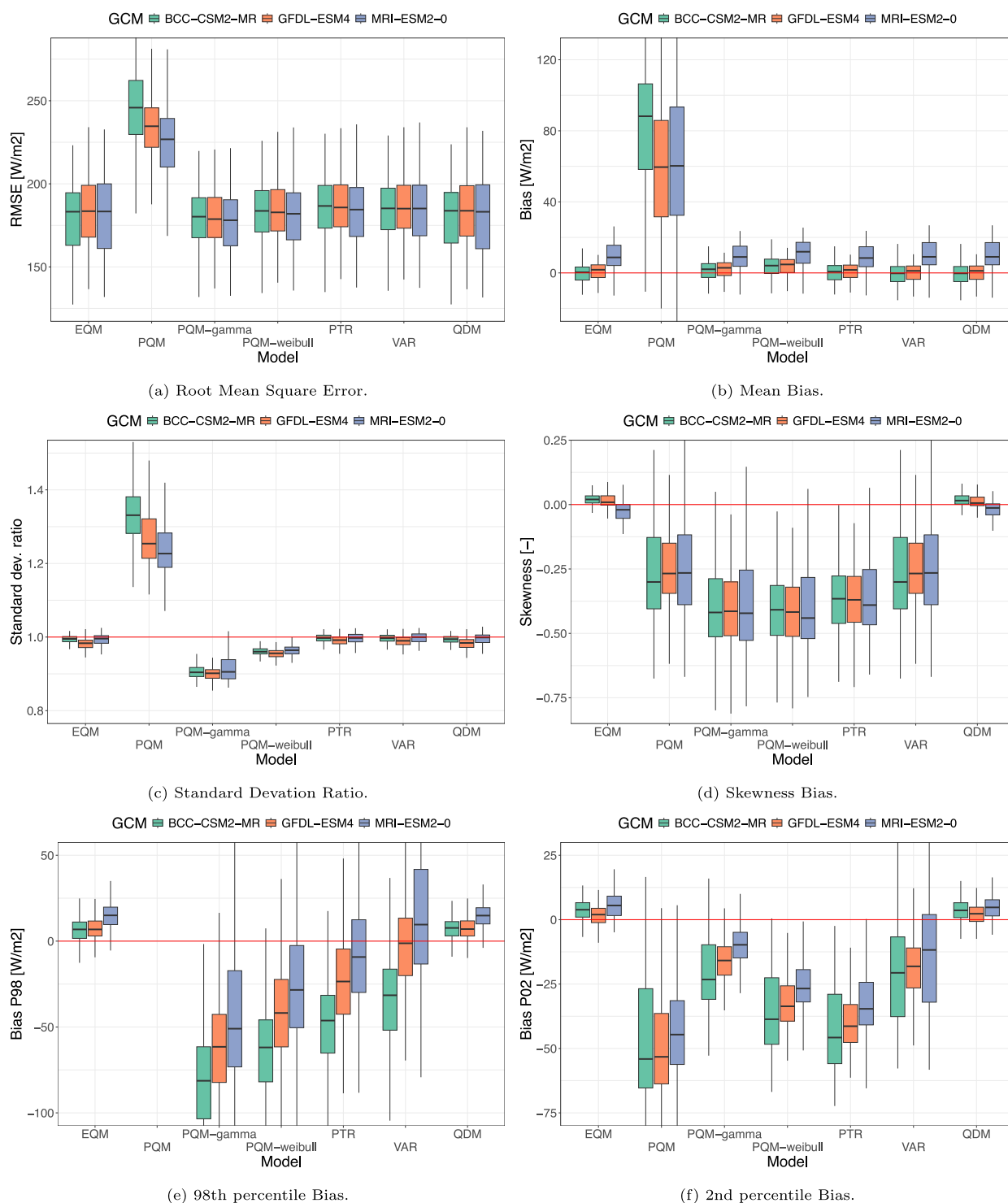


Fig. A.15. Validation metrics for solar radiation of the SD models for the three GCMs. Each boxplot represents the performance of an individual model for a given variable and metric. Missing points indicate cases where the SD model exhibited poor performance and was therefore excluded from further analysis. Abbreviations: AS-W (Additive Scaling), EQM and EQM (Empirical Quantile Mapping), PQM (Parametric Quantile Mapping), VAR (Variance Scaling), QDM (Quantile Delta Mapping), PTR (Power Transformation).

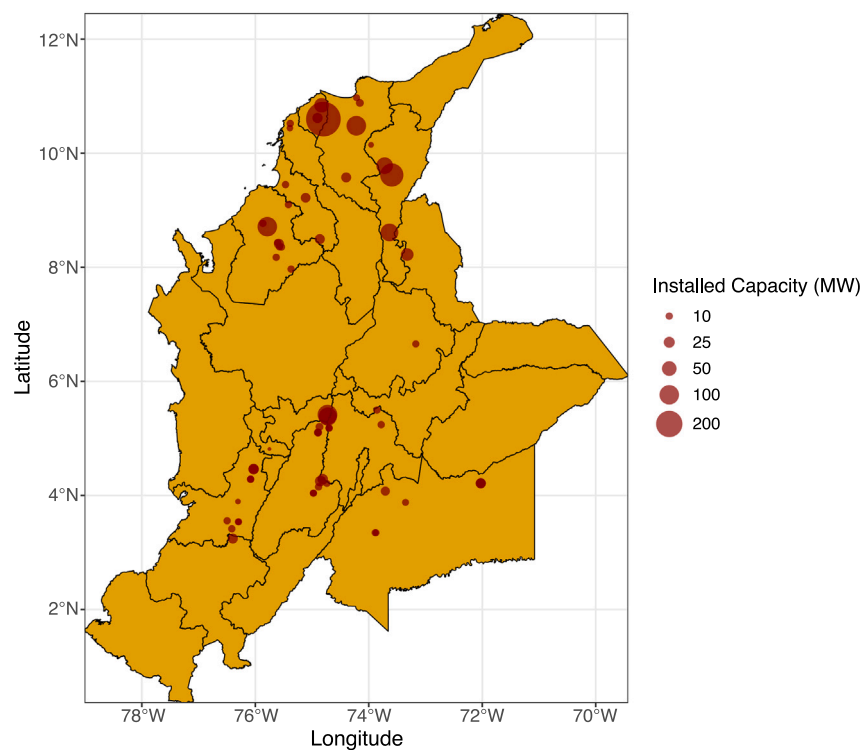


Fig. A.16. Spatial distribution of solar farms in Colombia based on official XM data. The size of each marker represents the installed photovoltaic capacity of the corresponding solar farm.

Appendix. Supplementary material

See Figs. A.13–A.16.

Data availability

Data will be made available on request.

References

- [1] H. Lee, J. Skea, P.R. Shukla, A. Revi, Y.N. Naimi, R.M. Reyes, C. Méndez, et al., Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Intergovernmental Panel on Climate Change (IPCC), Geneva, Switzerland, 2023, URL <https://www.ipcc.ch/report/sixth-assessment-report-cycle/>. ar6 Synthesis Report.
- [2] P.R. Shukla, J. Skea, R. Slade, A. Al Khourdajie, R. van Diemen, D. McCollum, M. Pathak, S. Some, P. Vyas, R. Fradera, M. Belkacemi, A. Hasija, G. Lisboa, S. Luz, J. Malley, Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, Cambridge, UK and New York, NY, USA, 2022, URL <https://www.ipcc.ch/report/ar6/wg3/>.
- [3] IRENA, Renewable Power Generation Costs in 2023, Tech. Rep., International Renewable Energy Agency (IRENA), Abu Dhabi, 2024, URL <https://www.irena.org/Publications/2024/Sep/Renewable-Power-Generation-Costs-in-2023>. (Accessed 27 June 2025).
- [4] S. Bouckaert, A.F. Pales, C. McGlade, U. Remme, B. Wanner, L. Varro, D. D'Ambrosio, T. Spencer, Net Zero by 2050: A Roadmap for the Global Energy Sector, Tech. Rep., International Energy Agency (IEA), 2021, URL <https://trid.trb.org/View/1856381>.
- [5] X. Zhao, G. Huang, C. Lu, X. Zhou, Y. Li, Impacts of climate change on photovoltaic energy potential: A case study of china, *Appl. Energy* 280 (2020) 115888.
- [6] M. Wild, Enlightening global dimming and brightening, *Bull. Am. Meteorol. Soc.* 93 (1) (2012) 27–37.
- [7] A. Sanchez-Lorenzo, A. Enriquez-Alonso, M. Wild, J. Trentmann, S.M. Vicente-Serrano, A. Sanchez-Romero, R. Posselt, M.Z. Hakuba, Trends in downward surface solar radiation from satellites and ground observations over europe during 1983–2010, *Remote Sens. Environ.* 189 (2017) 108–117.
- [8] J. Li, Y. Jiang, X. Xia, Y. Hu, Increase of surface solar irradiance across east china related to changes in aerosol properties during the past decade, *Environ. Res. Lett.* 13 (3) (2018) 034006.
- [9] B. Jahani, Y. Dinpashoh, M. Wild, Dimming in iran since the 2000s and the potential underlying causes, *Int. J. Climatol.* 38 (3) (2018) 1543–1559.
- [10] Z. Zhou, A. Lin, L. Wang, W. Qin, Y. Zhong, L. He, Trends in downward surface shortwave radiation from multi-source data over china during 1984–2015, *Int. J. Climatol.* 40 (7) (2020) 3467–3485.
- [11] I. Huber, L. Bugliaro, M. Ponater, H. Garny, C. Emde, B. Mayer, Do climate models project changes in solar resources? *Sol. Energy* 129 (2016) 65–84.
- [12] L. Yang, J. Jiang, T. Liu, Y. Li, Y. Zhou, X. Gao, Projections of future changes in solar radiation in china based on cmip5 climate models, *Glob. Energy Interconnect.* 1 (4) (2018) 452–459.
- [13] J. Müller, D. Folini, M. Wild, S. Pfenniger, Cmp-5 models project photovoltaics are a no-regrets investment in europe irrespective of climate change, *Energy* 171 (2019) 135–148.
- [14] Y. Dong, K.C. Armour, M.D. Zelinka, C. Proistosescu, D.S. Battisti, C. Zhou, T. Andrews, Intermodel spread in the pattern effect and its contribution to climate sensitivity in cmip5 and cmip6 models, *J. Clim.* 33 (18) (2020) 7755–7775.
- [15] M. Wild, The global energy balance as represented in cmip6 climate models, *Clim. Dyn.* 55 (3) (2020) 553–577.
- [16] R. Dutta, K. Chanda, R. Maity, Future of solar energy potential in a changing climate across the world: A cmip6 multi-model ensemble analysis, *Renew. Energy* 188 (2022) 819–829.
- [17] J. Guo, Z. Chen, J. Meng, H. Zheng, Y. Fan, L. Ji, X. Wang, X. Liang, Picturing china's photovoltaic energy future: Insights from cmip6 climate projections, *Renew. Sustain. Energy Rev.* 189 (2024) 114026.
- [18] R.C. Scheele, S. Fiedler, What drives historical and future changes in photovoltaic power production from the perspective of global warming? *Environ. Res. Lett.* 19 (1) (2023) 014030.
- [19] S. Baur, B.M. Sanderson, R. Séférian, L. Terray, Solar radiation modification challenges decarbonization with renewable solar energy, *Earth Syst. Dyn.* 15 (2) (2024) 307–322.
- [20] C.F. Zuluaga, A. Avila-Diaz, F.B. Justino, F.R. Martins, W.L. Ceron, The climate change perspective of photovoltaic power potential in Brazil, *Renew. Energy* 193 (2022) 1019–1031.
- [21] A. Isaza, M. Kay, J.P. Evans, A. Prasad, S. Bremner, Maximizing photovoltaic potential and minimizing costs in a future warmer climate: The role of atmospheric aerosols and greenhouse gas emissions, *Renew. Energy* 219 (2023) 119561.
- [22] G. Narvaez, M. Bressan, A. Pantoja, L.F. Giraldo, Climate change impact on photovoltaic power potential in south america, *Environ. Res. Commun.* 5 (8) (2023) 081004.

- [23] J. Vega Araújo, M. Muñoz Cabré, Energía solar y eólica en Colombia: Panorama y resumen de políticas 2022, 2023, <http://dx.doi.org/10.51414/sei2023.016>.
- [24] J.J.C. Eras, M.B. Morejón, A.S. Gutiérrez, A.P. García, M.C. Ulloa, F.J.R. Martínez, J.G. Rueda-Bayona, A look to the electricity generation from non-conventional renewable energy sources in Colombia, *Int. J. Energy Econ. Policy* 9 (1) (2019) 15–25.
- [25] Unidad de Planeación Minero Energética (UPME), Plan indicativo de expansión de generación 2025–2039, 2025, URL https://docs.upme.gov.co/SIMEC/Energia%20Electrica/PIEG/2025-2039/Plan_Generacion_2025-2039.pdf. (Accessed 01 May 2025).
- [26] EquipoAnaliticaXM, API XM, 2025, https://github.com/EquipoAnaliticaXM/API_XM.
- [27] G. Narvaez, L.F. Giraldo, M. Bressan, A. Pantoja, The impact of climate change on photovoltaic power potential in southwestern Colombia, *Heliyon* 8 (10) (2022).
- [28] H.A. Arregocés, R. Rojano, D. Pinto Pimienta, A comprehensive analysis of future solar energy potential variations using a cmip6 multi-model ensemble approach in Colombia, *Clean Energy* 8 (4) (2024) 79–89.
- [29] P.A. Arias, G. Ortega, L.D. Villegas, J.A. Martínez, Colombian climatology in CMIP5/CMIP6 models: Persistent biases and improvements, *Rev. Fac. Ing.* (100) (2021) 75–96, <http://dx.doi.org/10.17533/udea.redin.20210525>.
- [30] D. Maraun, M. Widmann, Statistical Downscaling and Bias Correction for Climate Research, Cambridge University Press, 2018, <http://dx.doi.org/10.1017/9781107588783>.
- [31] DANE, Proyecciones de población, 2020, URL <https://www.dane.gov.co/index.php/estadisticas-por-tema/demografia-y-poblacion/proyecciones-de-poblacion>.
- [32] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. De Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R.J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. de Rosnay, I. Rozum, F. Vamborg, S. Villaume, J. Thépaut, The ERA5 global reanalysis, *Q. J. R. Meteorol. Soc.* 146 (730) (2020) 1999–2049, <http://dx.doi.org/10.1002/qj.3803>.
- [33] Copernicus Climate Change Service, Climate Data Store, ERA5 hourly data on single levels from 1940 to present. copernicus climate change service (C3S) climate data store (CDS), 2023, <http://dx.doi.org/10.24381/cds.adbb2d47>, (Accessed 09 October 2024).
- [34] N. Lu, L. Yao, J. Qin, K. Yang, M. Wild, H. Jiang, High emission scenario substantially damages China's photovoltaic potential, *Geophys. Res. Lett.* 49 (20) (2022) e2022GL100068.
- [35] X. Chen, H. Mao, N. Cheng, L. Ma, Z. Tian, Y. Luo, C. Zhou, H. Li, Q. Wang, W. Kong, et al., Climate change impacts on global photovoltaic variability, *Appl. Energy* 374 (2024) 124087.
- [36] X. Costoya, M. DeCastro, D. Carvalho, M. Gómez-Gesteira, Assessing the complementarity of future hybrid wind and solar photovoltaic energy resources for North America, *Renew. Sustain. Energy Rev.* 173 (2023) 113101.
- [37] Y. Wang, X. Zhao, A. Mamtimin, H. Sayit, S. Abulizi, A. Maturdi, F. Yang, W. Huo, C. Zhou, X. Yang, et al., Evaluation of reanalysis datasets for solar radiation with in situ observations at a location over the gobi region of Xinjiang, China, *Remote. Sens.* 13 (21) (2021) 4191.
- [38] J.P. Dunne, L. Horowitz, A. Adcroft, P. Ginoux, I. Held, J. John, J.P. Krasting, S. Malyshev, V. Naik, F. Paulot, et al., The gfdl earth system model version 4.1 (gfdl-esm 4.1): Overall coupled model description and simulation characteristics, *J. Adv. Model. Earth Syst.* 12 (11) (2020) e2019MS002015.
- [39] S. Yukimoto, H. Kawai, T. Koshiro, N. Oshima, K. Yoshida, S. Urakawa, H. Tsujino, M. Deushi, T. Tanaka, M. Hosaka, et al., The meteorological research institute earth system model version 2.0, mri-esm2. 0: Description and basic evaluation of the physical component, *J. Meteorol. Soc. Jpn. Ser. II* 97 (5) (2019) 931–965.
- [40] T. Wu, Y. Lu, Y. Fang, X. Xin, L. Li, W. Li, W. Jie, J. Zhang, Y. Liu, L. Zhang, et al., The Beijing climate center climate system model (bcc-csm): The main progress from cmip5 to cmip6, *Geosci. Model. Dev.* 12 (4) (2019) 1573–1600.
- [41] IDEAM, Temperatura ambiente del aire - datos - portal nacional de datos abiertos, 2026, URL <https://www.datos.gov.co/Ambiente-y-Desarrollo-Sostenible/Temperatura-Ambiente-del-Aire/sbwg-7ju4>. (Accessed 10 October 2024).
- [42] M.J. Themeßl, A. Gobiet, G. Heinrich, Empirical-statistical downscaling and error correction of regional climate models and its impact on the climate change signal, *Clim. Change* 112 (2) (2012) 449–468, <http://dx.doi.org/10.1007/s10584-011-0224-4>.
- [43] J. Bedia, J. Baño-Medina, M.N. Legasa, M. Iturbide, R. Manzananas, S. Herrera, A. Casanueva, D. San-Martín, A.S. Cofiño, J.M. Gutiérrez, Statistical downscaling with the downscaler package (v3.1.0): Contribution to the VALUE intercomparison experiment, *Geosci. Model. Dev.* 13 (3) (2020) 1711–1735, <http://dx.doi.org/10.5194/gmd-13-1711-2020>.
- [44] M. Iturbide, J. Bedia, S. Herrera, J. Baño-Medina, J. Fernández, M. Frías, R. Manzananas, D. San-Martín, E. Cimadevilla, A. Cofiño, J. Gutiérrez, The R-based climate4R open framework for reproducible climate data access and post-processing, *Environ. Model. Softw.* 111 (2019) 42–54, <http://dx.doi.org/10.1016/j.envsoft.2018.09.009>.
- [45] J. Räisänen, O. Räty, Projections of daily mean temperature variability in the future: Cross-validation tests with ENSEMBLES regional climate simulations, *Clim. Dyn.* 41 (5–6) (2013) 1553–1568, <http://dx.doi.org/10.1007/s00382-012-1515-9>.
- [46] A. Amengual, V. Homar, R. Romero, S. Alonso, C. Ramis, A statistical adjustment of regional climate model outputs to local scales: Application to Platja de Palma, Spain, *J. Clim.* 25 (3) (2012) 939–957, <http://dx.doi.org/10.1175/JCLI-D-10-05024.1>.
- [47] C. Piani, J.O. Haerter, E. Coppola, Statistical bias correction for daily precipitation in regional climate models over Europe, *Theor. Appl. Climatol.* 99 (1–2) (2010) 187–192, <http://dx.doi.org/10.1007/s00704-009-0134-9>.
- [48] M.R. Tye, D.B. Stephenson, G.J. Holland, R.W. Katz, A Weibull approach for improving climate model projections of tropical cyclone wind-speed distributions, *J. Clim.* 27 (16) (2014) 6119–6133.
- [49] J. Chen, F.P. Brissette, R. Leconte, Uncertainty of downscaling method in quantifying the impact of climate change on hydrology, *J. Hydrol.* 401 (3–4) (2011) 190–202, <http://dx.doi.org/10.1016/j.jhydrol.2011.02.020>.
- [50] A.J. Cannon, S.R. Sobie, T.Q. Murdock, Bias correction of GCM precipitation by quantile mapping: How well do methods preserve changes in quantiles and extremes? *J. Clim.* 28 (17) (2015) 6938–6959, <http://dx.doi.org/10.1175/JCLI-D-14-00754.1>.
- [51] R. Leander, T.A. Buishand, Resampling of regional climate model output for the simulation of extreme river flows, *J. Hydrol.* 332 (3–4) (2007) 487–496, <http://dx.doi.org/10.1016/j.jhydrol.2006.08.006>.
- [52] R. Knutti, J. Sedláček, B.M. Sanderson, R. Lorenz, E.M. Fischer, V. Eyring, A climate model projection weighting scheme accounting for performance and interdependence, *Geophys. Res. Lett.* 44 (4) (2017) 1909–1918.
- [53] L. Brunner, A.G. Pendergrass, F. Lehner, A.L. Merrifield, R. Lorenz, R. Knutti, Reduced global warming from cmip6 projections when weighting models by performance and independence, *Earth Syst. Dyn.* 11 (4) (2020) 995–1012.
- [54] S. Jerez, I. Tobin, R. Vautard, J.P. Montávez, J.M. López-Romero, F. Thais, B. Bartok, O.B. Christensen, A. Colette, M. Déqué, et al., The impact of climate change on photovoltaic power generation in Europe, *Nat. Commun.* 6 (1) (2015) 10014.
- [55] W. Sawadogo, M.S. Reboita, A. Faye, R.P. Da Rocha, R.C. Odoulami, C.F. Olusegun, M.O. Adeniyi, B.J. Abiodun, M.B. Sylla, I. Diallo, et al., Current and future potential of solar and wind energy over Africa using the regcm4 cordex-core ensemble, *Clim. Dyn.* 57 (5) (2021) 1647–1672.